Jordan Normal Form

$T: V \to V$

$m_T(x) = (x-\lambda_1)^{e_1} \cdots (x-\lambda_k)^{e_k}$

$\Leftrightarrow$ $\exists \mathcal{B}$ a basis of V,

s.t. $(T)_{\mathcal{B}\mathcal{B}} = \begin{pmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ 0 & & J_r \end{pmatrix}$ $\to$ (block)

where $\forall i$, J_i is of the form $\begin{pmatrix} \lambda_{t_i} & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_{t_i} \end{pmatrix}$

Remark:

① 可能 $\exists$ some $J_i = \begin{pmatrix} \lambda_{t_i} & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_{t_i} \end{pmatrix}$ and $J_p = \begin{pmatrix} \lambda_{t_p} & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_{t_p} \end{pmatrix}$,

s.t. $\lambda_{t_i} = \lambda_{t_p}$

but the diag. entries λ_{t_i} must $\in \{\lambda_1, \cdots, \lambda_k\}$

② For each $\{\lambda_1, \cdots, \lambda_p, \cdots, \lambda_k\}$,

$\exists$ at least 1 Jordan block J_i, s.t. its diag. entry $\lambda_{t_i} = \lambda_p$

③ More precisely,

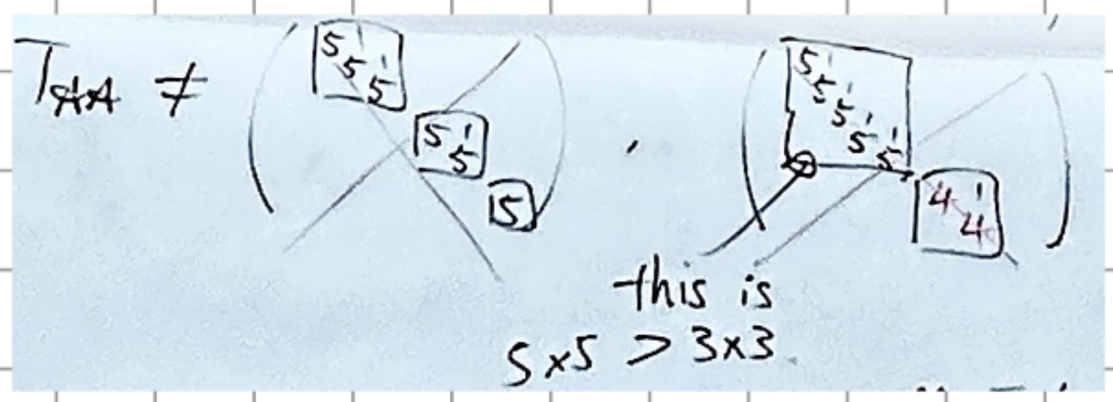
Let $J_{p_1}, \cdots, J_{p_x}$ be the blocks in $(T)_{\mathcal{B}\mathcal{B}}$

with diag. entry λ_p

Then the sizes of $J_{p_1}, \cdots, J_{p_x}$ are at most $e_p \times e_p$

(接上)

e.g., if $m_T(x) = (x-5)^3(x-4^2)$,

then $T_{AA} \neq$ 

④ In particular, if $\forall i, e_i = 1$

Then $\forall i$, J_i must be 1×1 block

Then $T_{AA} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_k \end{pmatrix}$ is diag.

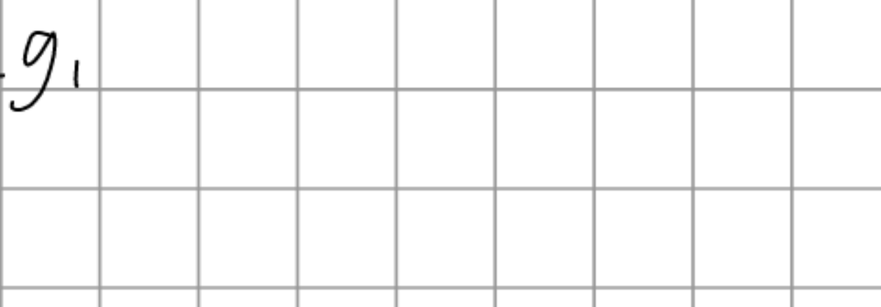
⑤ By changing the order of the basis $\mathcal{A}$, we can "move" the J_i's in $(T)_{\mathcal{AA}}$

e.g.

Note that (Section on matrix representations)

let $C = \ell_{\mathcal{AA}'}$,

$(T)_{\mathcal{A'A'}} = C^{-1}(T)_{\mathcal{AA}} C$

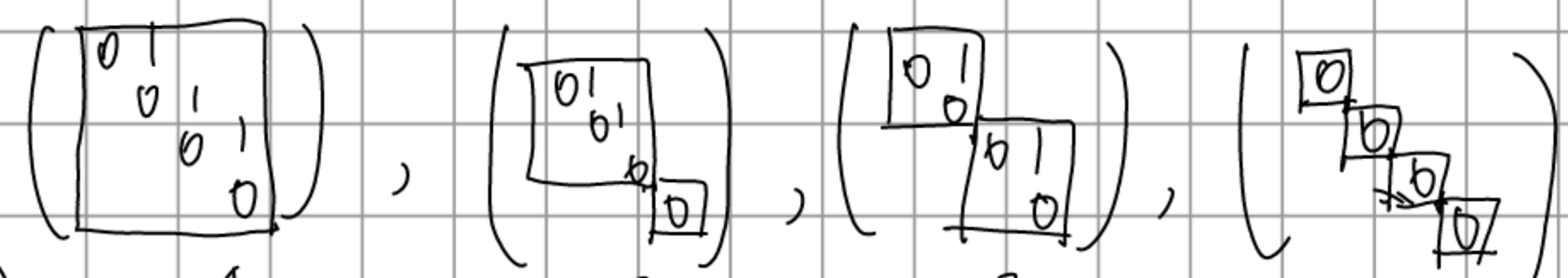
$\Rightarrow$  & are similar matrices

E.g.: Let $T: \mathbb{R}^4 \to \mathbb{R}^k$,

$\chi_T(x) = x^4$

C.-H. $\Rightarrow$ $m_T(x) = x$ or x^2 or x^3 or x^4

the possibilities of Jordan blocks



$m_T(x) =$ x^4 x^3 x^2 x

? ✓

Proof (of JNF):

(3 step: (1) $m_T(x) = x^m$ ... (2)

(1) (HW7) $\rightarrow$ $0 \subsetneq \ker(T) \subsetneq \ker(T^2) \subsetneq \cdots \subsetneq \ker(T^m) = V$

denote $\ker(T^i) = K_i$ $(K_0 = 0)$

Define $W_m = K_m / K_{m-1}$

Let $\mathcal{B}_m' = \{\underline{u}_1^m + K_{m-1}, \cdots, \underline{u}_i^m + K_{m-1}\}$

be a basis of W_m

Claim 1: $\mathcal{C}_m' := \{T(\underline{u}_1^m) + K_{m-2}, \cdots, T(\underline{u}_i^m) + K_{m-2}\}$

$\subseteq V/K_{m-2}$

① $\mathcal{C}_m' \subseteq W_{m-1} = K_{m-1}/K_{m-2}$

Proof (of Claim 1):

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Claim 2: $\mathcal{C}_m'$ is l. i. in W_{m-1}

Proof (

Let $\alpha_1(T(\underline{u}_1^m) + K_{m-2}) + \cdots + \alpha_i(T(\underline{u}_i^m) + K_{m-2}) = \underline{0}_{V/K_{m-2}}$

$\Leftrightarrow T(\alpha_1 \underline{u}_1^m + \cdots + \alpha_i \underline{u}_i^m) + K_{m-2} = \underline{0} + K_{m-2}$

$\Leftrightarrow T(\alpha_1 \underline{u}_1^m + \cdots + \alpha_i \underline{u}_i^m) \in K_{m-2} = \ker(T^{m-2})$

$\Leftrightarrow T^{m-2}(T(\alpha_1 \underline{u}_1^m + \cdots + \alpha_i \underline{u}_i^m)) = \underline{0}$

$\Leftrightarrow \alpha_1 \underline{u}_1^m + \cdots + \alpha_i \underline{u}_i^m \in \ker(T^{m-1}) = K_{m-1}$

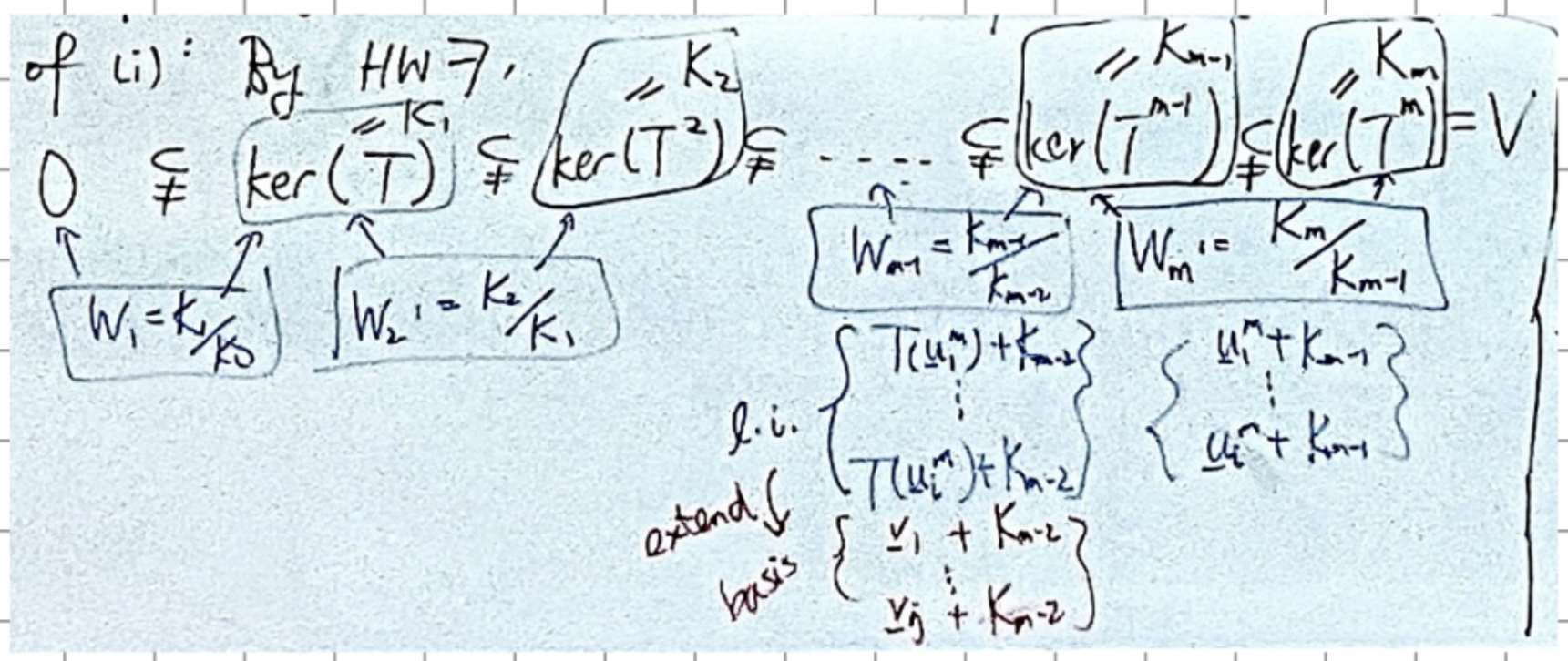
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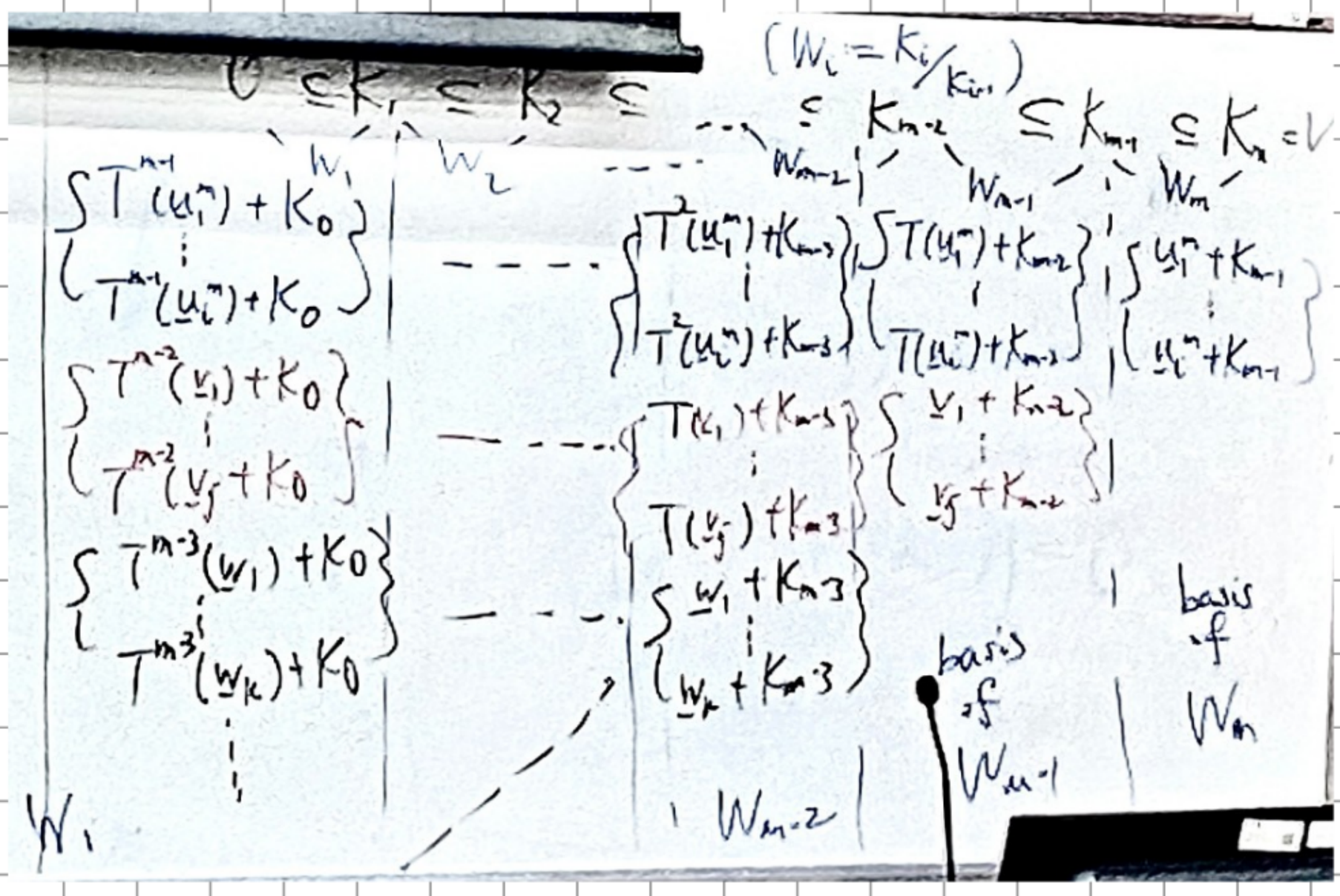
$\Leftrightarrow (\alpha_1 \underline{u}_1^m + \cdots + \alpha_i \underline{u}_i^m) + K_{m-1} = \underline{0} + K_{m-1}$

$\Leftrightarrow \alpha_1 (\underline{u}_1^m + K_{m-1}) + \cdots + \alpha_i (\underline{u}_i^m + K_{m-1}) = \underline{0}_{W_m}$

$\mathcal{B}_m'$ (basis of W_m)

$\Leftrightarrow \alpha_1 = \cdots = \alpha_i = 0$





Claim 3:

$$\mathcal{A} = \left\{ \begin{array}{l} T^{m-1}(\underline{u}_1^m), \cdots, T^2(\underline{u}_1^m), T(\underline{u}_1^m), \underline{u}_1^m, \\ \vdots \\ T^{m-1}(\underline{u}_i^m), \cdots, T^2(\underline{u}_i^m), T(\underline{u}_i^m), \underline{u}_i^m, \\ T^{m-2}(\underline{v}_1^m), \cdots, T^2(\underline{v}_1^m), T(\underline{v}_1^m), \underline{v}_1^m, \\ \vdots \\ T^{m-2}(\underline{v}_j^m), \cdots, T^2(\underline{v}_j^m), T(\underline{v}_j^m), \underline{v}_j^m, \\ T^{m-3}(\underline{w}_1^m), \cdots, T^2(\underline{w}_1^m), T(\underline{w}_1^m), \underline{w}_1^m, \\ \vdots \end{array} \right\}$$

Then $(T)_{\mathcal{A}\mathcal{A}}$ is given by

